

Lecture 4

Sampling Distributions

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LPO.8800: Statistical Methods for Education Research



Introduction



- Probability concepts (sample space, events, complements, $\cup$, $\cap$)
- Probability as relative frequency in the population
- Rules, e.g. $P(A \cup B)$ when A and B are mutually exclusive or not
- $P(A \cap B)$ when A and B are independent events
- Conditional probability, Bayes Rule
- Discrete probability distributions (e.g., Bernoulli, binomial)
- Continuous probability distributions (uniform, normal)
- Expectations and variance



Sampling distributions



- Probability distributions are used to describe the possible outcomes of a random variable, and their probabilities
- Statistics calculated from random draws from the population (such as the sample mean $\bar{x}$) **are also random variables**
- That is, they take on different values from one sample to the next—**sampling variation**



2

Sampling distributions

- As random variables, sample statistics (like $\bar{x}$) *also* have probability distributions that describe the relative frequency of its outcomes
- This probability distribution has central tendency (e.g., a mean or expected value) and variation (variance, standard deviation)
- The probability distribution of a sample statistic (like $\bar{x}$) is called a **sampling distribution**
- The sampling distribution is a *population* probability distribution since it characterizes all possible samples from a source population.



3

Consider a population for x consisting of 10 values:

0 1 3 3 5 7 7 7 8 10

- We can easily calculate the *population* mean $\mu = E(x) = 5.1$ and population standard deviation $\sigma = 3.08$
- Now imagine we don't observe this population, but take a random sample of $n = 2$ from it, with replacement, and calculate the sample mean ($\bar{x}$)

4



Sampling distributions: example

The realized value of $\bar{x}$ depends on the sample drawn:

- (3, 3): $\bar{x} = 3$
- (1, 8): $\bar{x} = 4.5$
- (8, 10): $\bar{x} = 9$

Now imagine *all possible samples* of $n = 2$ one could draw, and their associated values of $\bar{x}$. (There are 100 such combinations).

This is relatively easy to do since the population is small (10 outcomes) and the sample size is small ($n=2$). This list gets dramatically larger with bigger populations and ns .

5



Sampling distributions: example

A complete list of all possible samples of size $n = 2$ and their $\bar{x}$:

Sample	Mean	Sample	Mean	Sample	Mean	Sample	Mean	Sample	Mean
0,0	0.0	3,0	1.5	5,0	2.5	7,0	3.5	8,0	4.0
0,1	0.5	3,1	2.0	5,1	3.0	7,1	4.0	8,1	4.5
0,3	1.5	3,3	3.0	5,3	4.0	7,3	5.0	8,3	5.5
0,3	1.5	3,3	3.0	5,3	4.0	7,3	5.0	8,3	5.5
0,5	2.5	3,5	4.0	5,5	5.0	7,5	6.0	8,5	6.5
0,7	3.5	3,7	5.0	5,7	6.0	7,7	7.0	8,7	7.5
0,7	3.5	3,7	5.0	5,7	6.0	7,7	7.0	8,7	7.5
0,7	3.5	3,7	5.0	5,7	6.0	7,7	7.0	8,7	7.5
0,8	4.0	3,8	5.5	5,8	6.5	7,8	7.5	8,8	8.0
0,10	5.0	3,10	6.5	5,10	7.5	7,10	8.5	8,10	9.0

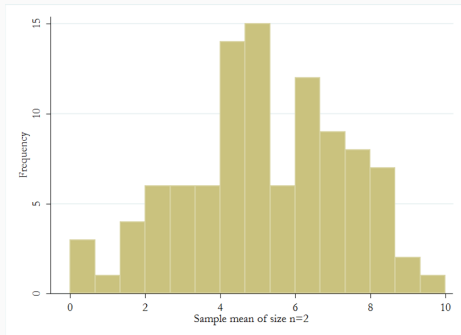
Sample	Mean	Sample	Mean	Sample	Mean	Sample	Mean	Sample	Mean
1,0	0.5	3,0	1.5	7,0	3.5	7,0	3.5	10,0	5.0
1,1	1.0	3,1	2.0	7,1	4.0	7,1	4.0	10,1	5.5
1,3	2.0	3,3	3.0	7,3	5.0	7,3	5.0	10,3	6.5
1,3	2.0	3,3	3.0	7,3	5.0	7,3	5.0	10,3	6.5
1,5	3.0	3,5	4.0	7,5	6.0	7,5	6.0	10,5	7.5
1,7	4.0	3,7	5.0	7,7	7.0	7,7	7.0	10,7	8.5
1,7	4.0	3,7	5.0	7,7	7.0	7,7	7.0	10,7	8.5
1,7	4.0	3,7	5.0	7,7	7.0	7,7	7.0	10,7	8.5
1,8	4.5	3,8	5.5	7,8	7.5	7,8	7.5	10,8	9.0
1,10	5.5	3,10	6.5	7,10	8.5	7,10	8.5	10,10	10.0

6



Sampling distributions: example

Plotting a histogram of these sample means shows the relative frequency (sampling distribution) of $\bar{x}$.

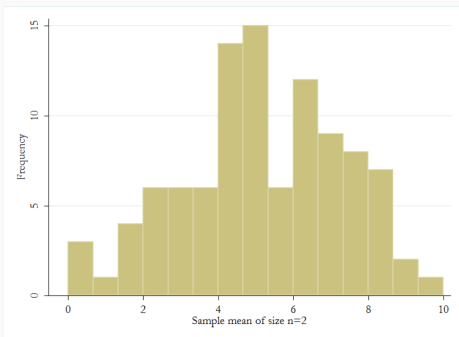


7



Sampling distributions: example

With a sample of $n = 2$, $\bar{x}$ ranges from 0 to 10. The mean of these 100 $\bar{x}$'s is 5.1 and standard deviation 2.19.



8



Sampling distributions: example

Now consider a simulation in which we *repeatedly* draw independent random samples of size $n = 8$ from the same population.

- Do this 50,000 times
- Calculate the sample mean each time
- Produce a histogram showing the relative frequency of these 50,000 sample means

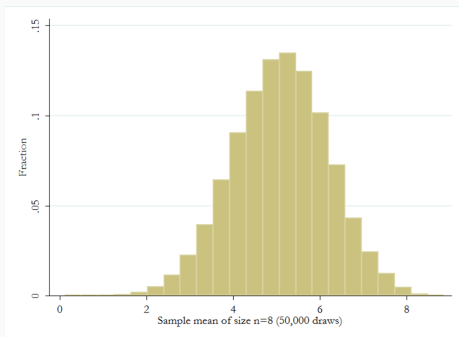
Note: syntax for conducting simulations in Stata is covered at the end of this lecture.

9



Sampling distributions: example

With a sample of $n = 8$, $\bar{x}$ is often between 3 and 7. The mean of these 50,000 $\bar{x}$'s is **5.1** and standard deviation **1.09**.



10



Standard error

- These sampling distributions allow us to see the probabilities that our sample statistic will take on certain values
- Ex: with a sample size of $n = 8$, how likely is it that we obtain a sample mean of 6.1 or higher? (About 16%)
- The standard deviation of a sampling distribution is called the **standard error**
- The standard error is a *population* quantity. It is a measure of variation in the statistic across many (repeated) samples.



The **standard error** of $\bar{x}$ represents how much the mean of a random sample of size n deviates, on average, from the population mean, over repeated samples.

It is $\sqrt{\text{Var}(\bar{x})}$

Denote $se(\bar{x})$ or $\sigma_{\bar{x}}$

12



Mean, variance, and standard error of $\bar{x}$

The population mean, variance, and standard error of $\bar{x}$ are easy to find, applying some basic rules of expectations and variance:

- The expectation of a sum of random variables is the sum of the expectations: $E(\sum_{i=1}^n x_i) = \sum_{i=1}^n E(x_i)$
- The variance of a sum of *independent* random variables is the sum of the variances: $\text{Var}(\sum_{i=1}^n x_i) = \sum_{i=1}^n \text{Var}(x_i)$
- With *independent* random variables x_i :
 $\text{Var}(\sum_{i=1}^n a x_i) = \sum_{i=1}^n a^2 \text{Var}(x_i)$

13



$\bar{x}$ is $\frac{1}{n} \sum_{i=1}^n x_i$, so:

$$E(\bar{x}) = E\left(\frac{1}{n} \sum_{i=1}^n x_i\right) = \frac{1}{n} \sum_{i=1}^n E(x_i) = \frac{1}{n} n\mu = \mu$$

$$\text{Var}(\bar{x}) = \text{Var}\left(\frac{1}{n} \sum_{i=1}^n x_i\right) = \left(\frac{1}{n}\right)^2 \sum_{i=1}^n \text{Var}(x_i) = \left(\frac{1}{n}\right)^2 n\sigma^2 = \frac{\sigma^2}{n}$$

$$\text{se}(\bar{x}) = \sqrt{\text{Var}(\bar{x})} = \frac{\sigma}{\sqrt{n}}$$

These apply no matter the underlying distribution of x .



Central limit theorem



Central limit theorem part 1

- The **central limit theorem** tells us something important about the sampling distribution of $\bar{x}$
- If x has a **normal** distribution with mean μ and standard deviation σ , i.e. $x \sim N(\mu, \sigma)$, then the sampling distribution of $\bar{x}$ is **normal** with a mean of μ and a standard error of $\sigma_{\bar{x}} = \sigma/\sqrt{n}$
 - $E(\bar{x}) = \mu$ and $Var(\bar{x}) = \sigma^2/n$
- Can write: $\bar{x} \sim N(\mu, \frac{\sigma}{\sqrt{n}})$
- Think about what happens as $n \rightarrow \infty$
 - **Law of Large Numbers:** as sample size grows to infinity, sampling variation shrinks to zero



15

Central limit theorem part 2

- Even if x does *not* have a normal distribution, with a large enough n , the sampling distribution of $\bar{x}$ is **approximately normal** with a mean of μ and a standard error of $\sigma_{\bar{x}} = \sigma/\sqrt{n}$
- This is a remarkable and very useful result. It helps us easily “quantify our uncertainty” about $\bar{x}$, since we can determine from the normal distribution the likelihood of $\bar{x}$ taking on certain values.
- As a rule of thumb, $n \geq 30$ is often thought of as “large” for the sampling distribution of $\bar{x}$



16

There is sampling variation in $\bar{x}$ for two reasons:

- **x itself varies in the population (σ).** Different random samples of size n will produce different answers. Holding n constant, the more variability in the population (σ), the more sampling variation.
- **The sample size.** A smaller sample (n) will result in greater sampling variation than a larger one. Holding σ constant, the more data that is used, the more *precise* the estimate.

Put another way, the standard error $\sigma_{\bar{x}}$ increases with σ and decreases with n .



17

Central limit theorem: example

Consider a population: all births in the state of Maryland in 1994. The population size is 73,971

- The outcome of interest (x) is the birth weight of a newborn
- Suppose we know from the state census of births that the true population mean and standard deviation for birth weight in this year are: $\mu = 3320\text{g}$ and $\sigma = 681.7\text{g}$
- Note: in the real world, we *don't know* population parameters (μ, σ), but rather are trying to estimate them. For the sake of illustration, let's pretend we know this information!



18

Central limit theorem: example

Now draw 1,000 samples of newborns at random, of size n :

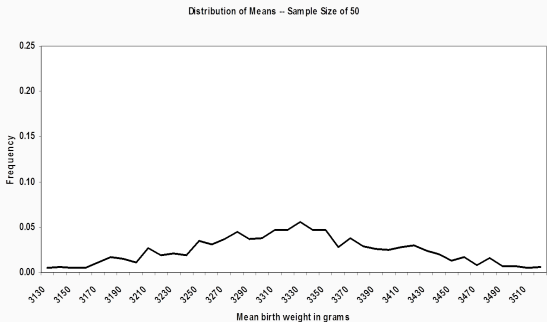
- $n = 50$ births
- $n = 100$ births
- $n = 200$ births
- $n = 1000$ births
- $n = 2000$ births

Plot the relative frequency of $\bar{x}$.



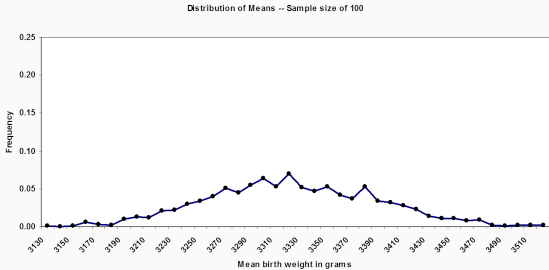
19

Central limit theorem: example



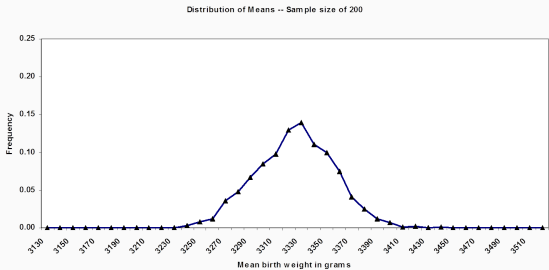
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Central limit theorem: example



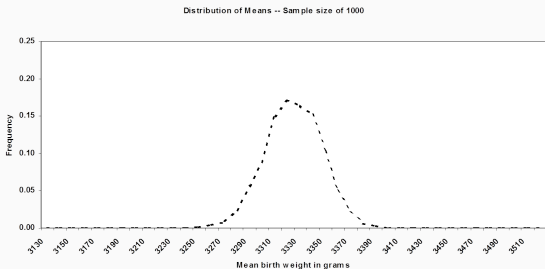
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Central limit theorem: example



22

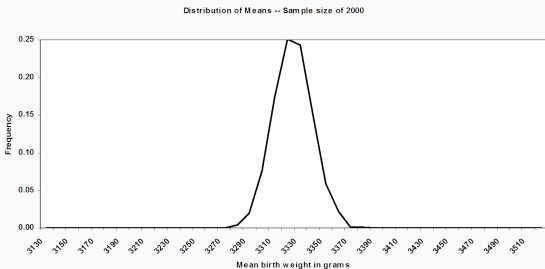
Central limit theorem: example



23

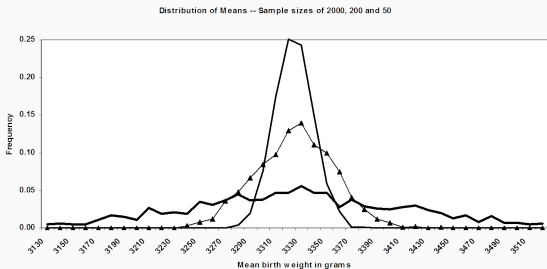


Central limit theorem: example



24





25



Central limit theorem: example

In this example where we know the population mean μ and σ^2 :

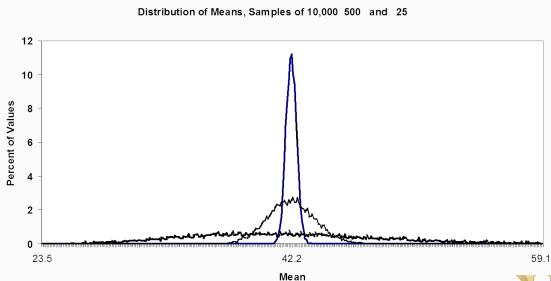
- $E(\bar{x}) = \mu = 3320$.
- $Var(\bar{x}) = \sigma^2/n = 681.7^2/n$
- $se(\bar{x}) = \sqrt{Var(\bar{x})} = \sigma/\sqrt{n} = 681.7/\sqrt{n}$

26



Central limit theorem: example

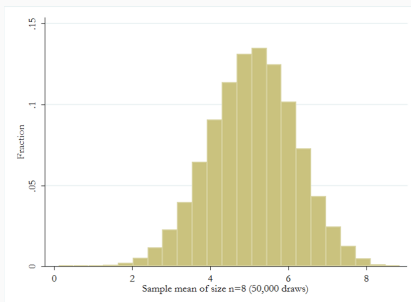
Another example using random samples of household income from the 1990 Census:



27

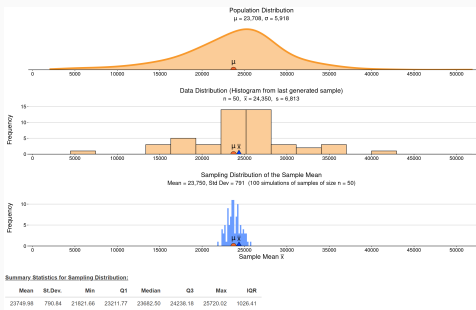
Central limit theorem: example

In the first simulation x was not even close to normally distributed, and n was only 8. Yet the sampling distribution of $\bar{x}$ looks very normal:



28

See the Github site for links to some great web apps for visualizing sampling distributions and the CLT. For example:



Applications

- We sample x from a *normally distributed population* in which $\mu = 15$ and $\sigma = 3$
- Our sample size is $n = 16$
- What will the sampling distribution of $\bar{x}$ look like?



30

Application 1

From the CLT we know that $\bar{x}$:

- Has a **normal** distribution
- Has a mean of 15
- Has a standard error of $\sigma/\sqrt{n} = 3/\sqrt{16} = 0.75$

Critically, we did *not* have to actually take repeated samples to know this!

In practice, we don't know the specific values of μ or σ but we do know generally that $E(\bar{x}) = \mu$ and $se(\bar{x}) = \sigma/\sqrt{n}$, and that $\bar{x}$ has a normal distribution.



31

Application 2[a]

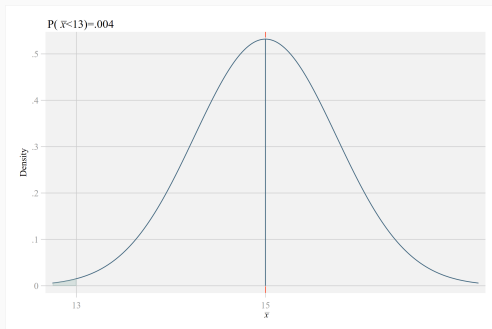
Using the same information from Application 1, what is the probability that a random sample of $n = 16$ is drawn with an $\bar{x}$ of 13 or less?

- We know that $\bar{x} \sim N(15, 0.75)$
- So $z = \frac{\bar{x}-15}{0.75} \sim N(0, 1)$
- $Pr(\bar{x} \leq 13) = Pr\left(\frac{\bar{x}-15}{0.75} \leq \frac{13-15}{0.75}\right) = Pr(z < -2.67) = 0.0038$
- Stata: `display normal(-2.67)`

32



Application 2[a]



Note: see Github for Stata graph syntax

33



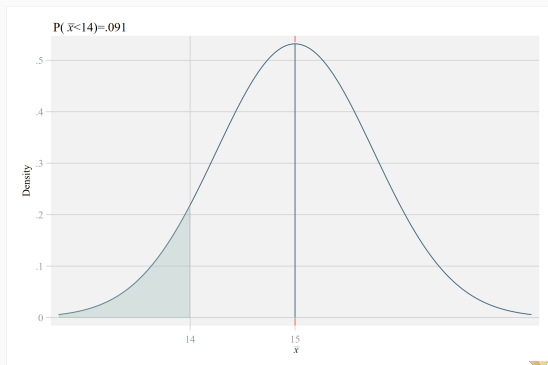
What is the probability that one would draw a random sample of $n = 16$ and obtain an $\bar{x}$ of **14 or less**?

- $Pr(\bar{x} \leq 14) = Pr\left(\frac{\bar{x} - 15}{0.75} \leq \frac{14 - 15}{0.75}\right) = Pr(z < -1.33) = 0.0918$
- Stata: `display normal(-1.33)`

34



Application 2[b]



35



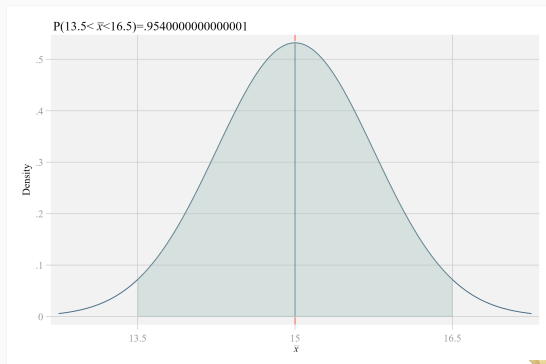
What is the probability that one would draw a random sample of $n = 16$ and obtain an $\bar{x}$ **between 13.5 and 16.5**?

- The z-values for 13.5 and 16.5 are:
- $\frac{(13.5-15)}{0.75} = -2$
- $\frac{(16.5-15)}{0.75} = +2$
- $Pr(-2 \leq z \leq 2) \approx 0.95$
- Stata: `display normal(2) - normal(-2) = 0.954`

36



Application 2[c]



37



Use the CLT to **determine the necessary sample size** for a given amount of precision.

- Suppose we know that family income in 1997 averaged \$32K with a standard deviation of \$17K (μ and σ)
- We plan to survey families about their income, and do not want $\bar{x}$ to vary too much
- How many families do we have to survey so that $\bar{x}$ is within \$4K of μ 95% of the time?



- We know that family income $\mu = 32$ and $\sigma = 17$
- By the CLT, $\bar{x} \sim N(32, 17/\sqrt{n})$
- Find the n so that $Pr(28 \leq \bar{x} \leq 36) = 0.95$
- In other words, $\bar{x}$ can be expected to fall between 28 and 36 (\$4K from μ) 95% of the time.



Expressing using z-scores:

$$P\left(\frac{28 - 32}{17/\sqrt{n}} < z < \frac{36 - 32}{17/\sqrt{n}}\right) = 0.95$$

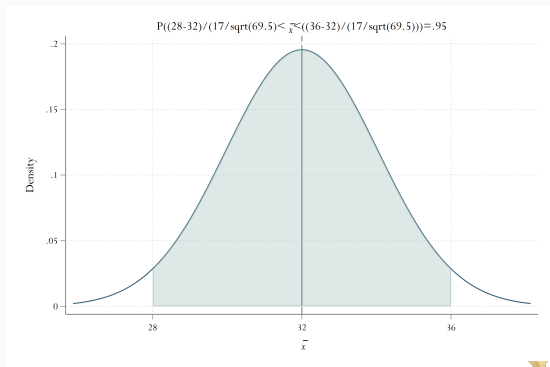
We know the probability that z falls between -1.96 and 1.96 is 0.95. So:

$$\left(\frac{36 - 32}{17/\sqrt{n}} = 1.96\right) \text{ or } n = 69.4$$

40



Application 3



41



A smaller n implies greater sampling variability (less precision). It follows that small samples are more likely to exhibit extreme values than large samples.

- Kane & Staiger (2002): *The Promise and Pitfalls of Using Imprecise School Accountability Measures*
- School accountability systems often reward the very “best” schools and sanction the very “worst” (e.g., top and bottom 5%)
- Under *No Child Left Behind* schools could be penalized based on poor subgroup performance



42

Precision and school accountability

From Kane & Staiger (2002):

Table 1
Awards and Sanctions Among Elementary Schools in North Carolina

<i>School Size</i>	<i>Percentage Ever “Top 25” 1997–2000</i>	<i>Percentage Ever Assigned Assist. Team 1997–2000</i>	<i>Mean Gain in Math</i>	<i>Between-School Variance in Mean Gain in Math</i>
Smallest decile	27.7%	1.2%	.020	.048
2nd	11.8	4.7	–.007	.030
3rd	8.2	7.1	.008	.028
4th	3.6	1.2	.009	.026
5th	2.4	0	–.002	.024
6th	3.6	0	.019	.018
7th	4.8	0	.007	.016
8th	7.1	0	.006	.016
9th	0	1.2	–.007	.015
Largest decile	1.2	0	–.011	.011
Total	7.0	1.5	.004	.023

Note: The above refers to the 840 regular public elementary schools for whom we had data from 1994 through 2000. Charter schools are not included.

43



From Kane & Staiger (2002):

This pattern of test scores by school size poses some challenges for designing a system of rewards and sanctions. If the state requires schools to exceed a high threshold in order to receive a reward, there will be a much higher chance that the small school will win the award than a large school—even if the small and large schools have the same “true” performance level. Indeed, large schools are likely to be unmotivated by school accountability programs that emphasize the extremes, since they have very little chance of ever appearing in the tails of the distribution. While they may not be fully cognizant of the statistical issues involved, those working at schools of varying sizes would be expected to learn over time that their efforts have a greater or lesser impact on their likelihood of winning.



Estimators



Because $\bar{x}$ was used with a random sample to estimate μ , it is called an **estimator**. Think of an estimator as a formula, “recipe,” or procedure used to estimate a population parameter.

- What makes $\bar{x}$ a *good* estimator of μ ?
- Is $\bar{x}$ the *best* estimator of μ ?



- An **unbiased estimator** of a population parameter (like μ , for example) has a sampling distribution with a mean equal to the parameter being estimated.
- In other words, *in repeated samples* the average realized value of the estimator is the parameter value you're looking for



- The CLT tells us that the mean of the sampling distribution for $\bar{x}$ is μ . So, $\bar{x}$ is an *unbiased* estimator of μ . $E(\bar{x}) = \mu$
- When asking whether an estimator is “biased” you are asking whether there is anything that causes it to *systematically* over or under-estimate the true population quantity.
- Critical point: unbiased estimators are not necessarily *correct* in any specific sample. In fact, they probably won’t be! They just use a *procedure* that will, on average, yield the correct estimate.



47

Estimators

- In many applications of statistics there are multiple estimators available for estimating a population parameter.
- How can we choose among these? Which estimator is the “best?”
- Unbiasedness is one good quality of an estimator.
- **Efficiency** is another. One estimator $\tilde{x}$ is more *efficient* than another $\bar{x}$ if the variance of the sampling distribution of $\tilde{x}$ is less than the variance of the sampling distribution of $\bar{x}$.
 - We’ve seen one example of this already: $\bar{x}$ estimated with $n = 100$ is more efficient than $\bar{x}$ estimated with $n = 10$.



48

- As noted, an **estimator** is a procedure for estimating a population parameter (e.g., $\bar{x}$).
- The population quantity one is trying to estimate is called the **estimand** (e.g., μ).
- The numerical result obtained from a given sample is called an **estimate**.

In any given application, it is useful to remember:

$$\text{Estimate} = \text{Estimand} + \text{Bias} + \text{Noise}$$



- Bias is *systematic* error
- Noise is *idiosyncratic* error (e.g., sampling variation)



Sampling distributions for proportions



Sampling distributions for proportions

Consider a Bernoulli random variable x . We know the mean of x represents the *proportion* of cases in which $x = 1$.

- $E(x) = \pi$, the proportion of cases in the *population* where $x = 1$. (π is just a special name for μ whenever x is dichotomous).
- $1 - \pi$ is the proportion of cases in the population where $x = 0$.
- $\hat{\pi}$ is the *sample* proportion of cases where $x = 1$.
- $\hat{\pi}$ is an estimator of π



When x is a Bernoulli variable, the population standard deviation of x is: $\sqrt{\pi(1-\pi)}$ (see Lecture 3). Applying the CLT:

- The sampling distribution of $\hat{\pi}$ is **approximately normal** when n is sufficiently large.
- Common rule of thumb: when $n\pi \geq 10$ and $n(1-\pi) \geq 10$
- This is a good example of the CLT when the underlying variable x does *not* have a normal distribution
- The mean of $\hat{\pi}$ will be π , and its standard error $\sqrt{\frac{\pi(1-\pi)}{n}}$
- $\hat{\pi} \sim N\left(\pi, \sqrt{\frac{\pi(1-\pi)}{n}}\right)$



52

Application 4

Suppose we know that 46% of the voting age population approves of the way Joe Biden is handling his job as president. If 100 adults are polled at random and asked whether or not they approve of Biden, the sample proportion in favor is $\hat{\pi}$.

- The sampling distribution of $\hat{\pi}$ is **approximately normal**
- With a mean of $\pi = 0.46$ (the population proportion)
- And a standard error of $\sigma_{\hat{\pi}} = \sqrt{\frac{\pi(1-\pi)}{n}} = \sqrt{\frac{0.46(1-0.46)}{100}} = 0.0498$
- Notice the notation $\sigma_{\hat{\pi}}$ for the standard error of the estimator $\hat{\pi}$

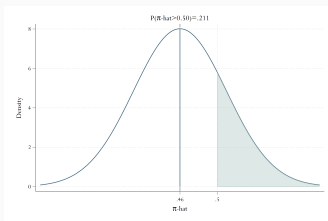


53

Application 4

What is the probability in a random sample of 100 adults that a *majority* approve of the way Joe Biden is handling his job as president?

- $Pr(\hat{\pi} > 0.50) = Pr\left(\frac{\hat{\pi} - 0.46}{0.0498} > \frac{0.50 - 0.46}{0.0498}\right) = Pr(z > 0.803)$
- In Stata: `1-normal(0.803) = 0.211`



54



Application 5

Continuing with the same example, how many adults would we need to sample so that $\hat{\pi}$ is within 2 points (0.02) of $\pi = 0.46$ 95% of the time?

- By the CLT, $\hat{\pi} \sim N\left(0.46, \sqrt{\frac{0.46(1-0.46)}{n}}\right)$
- Find the n so that $Pr(0.44 \leq \hat{\pi} \leq 0.48) = 0.95$
- In other words, $\hat{\pi}$ can be expected to fall between 0.44 and 0.48 95% of the time.

55



Expressing using z-scores:

$$P\left(\frac{0.44 - 0.46}{\sqrt{\frac{0.46(1-0.46)}{n}}} < z < \frac{0.48 - 0.46}{\sqrt{\frac{0.46(1-0.46)}{n}}}\right) = 0.95$$

We know the probability that z falls between -1.96 and 1.96 is 0.95 . So:

$$\left(\frac{0.48 - 0.46}{\sqrt{\frac{0.46(1-0.46)}{n}}} = 1.96\right) \text{ or } n = 2,385$$



56

Sampling distributions

You may have noticed something unusual about the examples in this lecture:

- Describing the sampling distribution of $\bar{x}$ requires knowing what σ is in the population.
- Describing the sampling distribution of $\hat{\pi}$ requires knowing what π is in the population.
- *If we have this information about the population, why are we taking a sample to begin with?*
- Good question. When the population quantities are unknown, they can be estimated. The standard error we use in practice is an *estimated* standard error, not the population standard error (see Lecture 5).



57

Simulating sampling distributions



Simulating sampling distributions

Simulation: using a random process that we have *control over* to produce data we can evaluate. Usually an iterative process whereby we draw multiple random samples (often 1000s). AKA Monte Carlo simulation.

What are the benefits of simulation?

- We define the process generating our data. Real data can be messy.
- We decide which assumptions hold and which don't, allowing us to test sensitivity to assumptions.
- Simulation is great for understanding what an estimator is doing, and the sampling distribution of that estimator.
- Some estimators have a complicated sampling distribution without simple expressions for their features. (Example: multi-step estimators, clustered data)



Typical steps:

1. Decide on the data generating process (DGP).
2. Use random draws from this DGP to produce a sample of size n .
3. Apply estimator to sample.
4. Store resulting estimate.
5. Iterate over steps 2-4.
6. Look at the resulting distribution of estimates.

We might be interested in the *mean* of the sampling distribution (e.g., is our estimator unbiased?), its *variance/standard error*, its quantiles, the prevalence of outlier values, etc.



59

Simulating sampling distributions in Stata

Refer to the handout “Useful Stata commands for simulation” on Github.

- Loops
- `simulate` command
- Bootstrapping
- `postfile` and related commands

As noted in Lecture 3, it's good practice to `set seed` before doing simulations.



60

`simulate` command example: first write a little program that draws a random sample and calculates statistics, then replicate it many times.

```
// Method 1 - program and simulate
clear
capture program drop app1

program app1, rclass
    version 15.1
    drop _all
    set obs 16
    gen x=rnormal(15, 3)
    summ x
    return scalar mean=r(mean)
    return scalar Var =r(Var)
end

set seed 4321
simulate mean=r(mean) var=r(Var), reps(1000) nodots: app1
summ mean var
```

61



Simulating sampling distributions in Stata

Same thing using `postfile` and related commands:

```
// Method 2 - postfile commands
// tempname is temporary holding place where results are held
// tempfile is temporary file for collecting results
clear
tempname results
tempfile meantable

// postfile tells Stata where results will be iteratively saved, which items
// will be saved, and where the final results will be collected
postfile `results' xmean xvar using `meantable'

forvalues j=1/1000 {
    clear
    drawnorm x, n(16) mean(15) sds(3)
    quietly summ
    post `results' (r(mean)) (r(Var))
}
postclose `results'

use `meantable', clear
summ xmean xvar
```

62



Bootstrapping involves repeated sampling *with replacement*. Recall the example that drew 50,000 samples of size $n = 8$ from a population. The following code repeatedly samples 8 observations from the data, executes `summarize` on variable `x1`, and retains the result stored in `r(mean)`:

```
bootstrap r(mean), reps(50000) size(8) saving(draws50k,  
replace): summ x1
```

After 50,000 replications, the results are saved in a file called *draws50k*. `bootstrap` is a prefix that comes before the main Stata command (`summ` in this case).

Useful when sampling from existing data.



Next time



Statistical inference: estimation

- Confidence intervals for means and proportions
- t-distribution
- Minimum sample size calculations
- Some more simulations!

